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APPENDIX 1

SOLVED EXAMPLE UNIT-1

1. Find the dimensions of a and b in the formula $[P + \frac{a}{V^2}][V - b] = RT$ where P is pressure and V is the volume of the gas

Solution:

By the principle of homogeneity, a/V^2 is of the dimensions of pressure and b is of the dimensions of volume.

$[a] = [\text{pressure}]\ [V^2] = [ML^{-1}T^{-2}]\ [L^6]$
$= [ML^5T^{-2}]$
$[b] = [V] = L^3$

2. Show that $(P^{-5/6}\rho^{1/2}E^{1/3})$ is of the dimension of time. Here P is the pressure, ρ is the density and E is the energy of a bubble)

Solution:

Dimension of Pressure $= [ML^{-1}T^{-2}]$

Dimension of density $= [ML^{-3}]$

Dimension of Energy $= [ML^2T^{-2}]$

By substituting in the given equation,

$= [ML^{-1}T^{-2}]^{-5/6}[ML^{-3}]^{1/2}[ML^2T^{-2}]^{1/3}$
$= M^{-5/6+1/2+1/3}L^{5/6-3/2+2/3}T^{5/3-2/3}$
$= M^0L^0T^1 = [T]$

3. Find the dimensions of mass in terms of Energy, length and time

Solution:

Let the dimensions of Energy, Length and Time be $[E],[L],[T]$ respectively.

We know that Force = mass x acceleration

$$\text{Mass} = \frac{Force}{acceleration}$$
$$= \frac{Workdone\ (or)\ Energy}{acceleration \times displacement}$$
$$[m] = \frac{Energy}{[acceleration][displacement]}$$
$$= \frac{[E]}{[LT^{-2}][L]} = \frac{[E]}{L^2T^{-2}} = [EL^{-2}T^2]$$

4. A physical quantity Q is found to depend on quantities x,y,z obeying